



ICAIN-2026 International Conference on Artificial Intelligence and Networking

26th -28th November 2026

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***** CALL FOR PAPERS *****

SPECIAL SESSION ON

AI and Networked Sensing for Climate Resilience, Agriculture and Earth Observation

SESSION ORGANIZERS:

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SESSION DESCRIPTION:

Climate volatility, water stress and food-system fragility are among the defining challenges of the coming decades, and nowhere more acutely than across the Global South. The sensing infrastructure needed to observe these systems has expanded dramatically: open satellite archives, dense low-cost ground sensor networks, agricultural UAVs, LPWAN telemetry and citizen-science platforms now generate environmental data at unprecedented volume and cadence. The binding constraint is no longer measurement but intelligence, and the networking that makes it usable. Observations arrive multi-modal, unevenly labelled, spatially and temporally misaligned, and are frequently collected in rural or remote settings where bandwidth, power and connectivity are severely limited. Turning such data into timely, trustworthy and actionable decisions is fundamentally a joint problem of artificial intelligence and network design.

This special session invites original, unpublished research spanning the full pipeline, from sensing and connectivity through learning to deployed decision support. Contributions are welcome on foundation models and deep learning for remote sensing, multi-source data fusion, edge and TinyML inference on constrained field devices, network architectures for

agricultural and environmental telemetry, and early-warning systems for floods, droughts and other extreme events. Work confronting the practical obstacles of the domain, including label scarcity, domain shift across geographies, intermittent connectivity, model reliability under distribution change, and the reproducibility of Earth-observation pipelines, is particularly encouraged, as are papers reporting field deployments, benchmarks and open datasets.

The session addresses artificial intelligence applied to environmental, agricultural and climate systems, and thereby complements rather than duplicates existing ICAIN-2026 sessions concerned with the energy efficiency of computing itself or with societal applications more broadly. It seeks to convene researchers from machine learning, remote sensing, wireless and sensor networking, agricultural engineering and climate science, and to move promising methods from publication toward field-validated impact.

RECOMMENDED TOPICS:

Topics to be discussed in this special session include (but are not limited to) the following:

- Deep learning and foundation models for satellite, multispectral and hyperspectral remote sensing
- Crop health monitoring, yield forecasting, pest and disease detection from UAV and satellite imagery
- Precision agriculture: AI-driven irrigation scheduling, nutrient management and variable-rate application
- Wireless sensor networks, LPWAN and IoT architectures for agricultural and environmental telemetry
- Edge intelligence and TinyML on low-power field devices under intermittent rural connectivity
- AI for weather nowcasting, extreme-event prediction and flood, drought and cyclone early-warning systems
- Multi-source data fusion across satellite, aerial, ground-sensor and citizen-science observations
- Land-use and land-cover change, deforestation, desertification and biodiversity monitoring
- Water resource management, groundwater assessment and irrigation-network optimisation
- Soil health, carbon-stock estimation and AI-assisted measurement, reporting and verification (MRV)
- Digital twins of farms, watersheds and regional climate systems; scenario modelling for adaptation
- Federated and privacy-preserving learning across farms and agencies; data sovereignty, equity and access
- Datasets, benchmarks, label scarcity, self-supervised learning and reproducibility in Earth observation

SUBMISSION PROCEDURE:

Researchers and practitioners are invited to submit papers for this particular theme session on **AI and Networked Sensing for Climate Resilience, Agriculture and Earth Observation** *on or before 30th August 2026*. All submissions must be original and may not be under review by another publication. INTERESTED AUTHORS SHOULD CONSULT THE CONFERENCE'S GUIDELINES FOR MANUSCRIPT SUBMISSIONS at <https://www.icain-conf.com/downloads>. All submitted papers will be reviewed on a double-blind, peer-review basis.

NOTE: While submitting a paper in this special session, please specify **AI and Networked Sensing for Climate Resilience, Agriculture and Earth Observation** at the top (above paper title) of the first page of your paper.



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